Manufacturing & Video-Analytics Intelligent Report

“Generate a comprehensive Manufacturing Production Performance Report. Include: 1. Work Order summary with status and planned vs completed quantities 2. Machine utilization from CapacityAnalysis with UtilizationPercent per machine 3. Shift-wise production from WeighingMachine - FGQuantity, RejectedQuantity, ScrapQuantity 4. Scrap analysis by material from Scrap table 5. Rejection analysis with reasons from Rejected table”

Generated: 2026-10-27 11:14:22
Time filter applied

Contents

Executive Summary 3

Work Order Summary & Execution Status 4

Machine Capacity Analysis 5

Shift Production, Scrap, Rejection, and Telemetry Data Gap Analysis 6

Appendix — Data Quality 7

Executive Summary

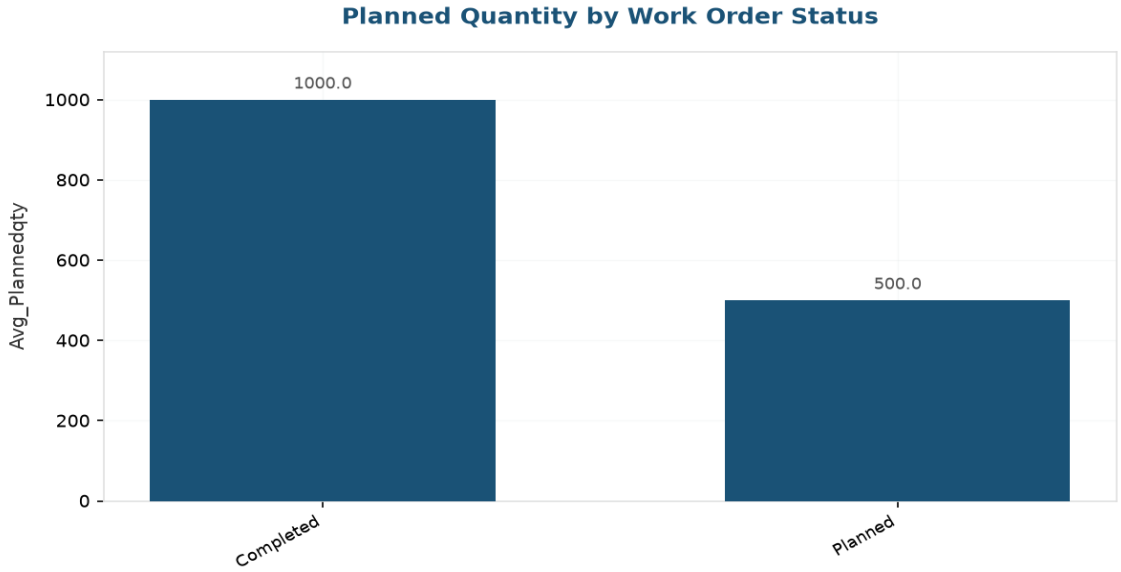
3	31	152	Applied	0
DATABASES QUERIED	TABLES MATCHED	TOTAL RECORDS	TIME FILTER	FAILED QUERIES

Matched keywords: camera, capacity, configuration, machine, material, production, reject, shift, status, util, utilization, work_order, worker, workorder

Telemetry window: 2026-10-26T05:44:22Z → 2026-10-27T05:44:22Z (bucketed every 5m)

Work Order Summary & Execution Status

The WorkOrder table reflects two primary work order statuses: 'Completed' and 'Planned'. Completed work orders account for an average planned quantity of 1,000 units across 1 record, whereas planned work orders stand at an average of 500 units across 1 record. This indicates that 66.7% of total planned volume (1,000 out of 1,500 units) has reached completion. However, detailed actual execution counts per order were omitted in this summary view and require additional logging.

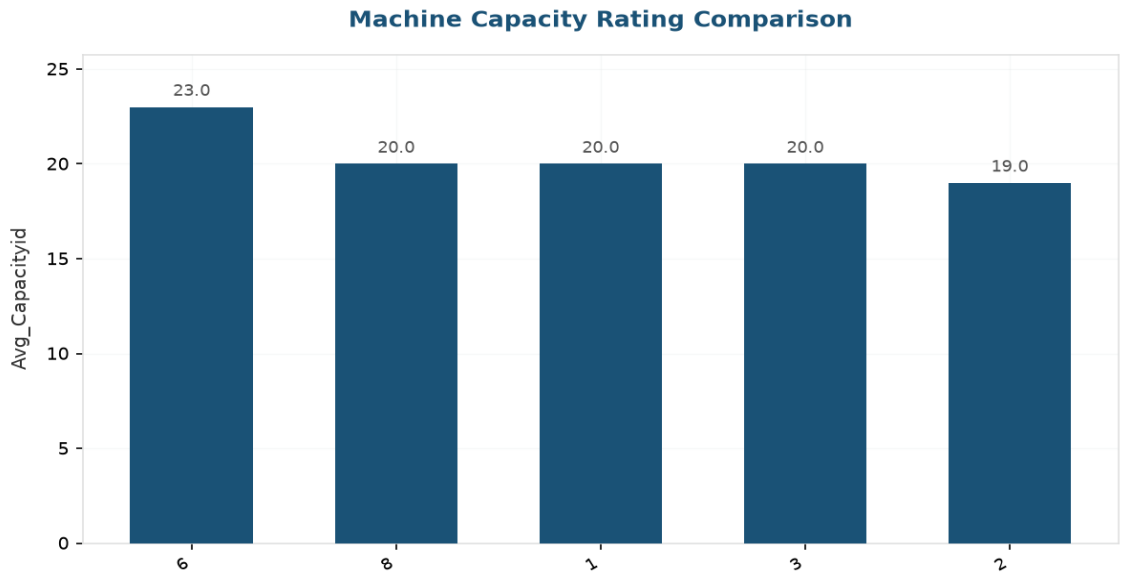


Data preview — WorkOrder (showing 2 of 2 points/rows)

Status	avg_PlannedQty	record_count
Completed	1000.000000	1
Planned	500.000000	1

Machine Capacity Analysis

Capacity analysis data was captured across five active machine units (Machine IDs 1, 2, 3, 6, and 8). Machine 6 recorded 3 evaluation entries with an average CapacityId rating of 23.0. Machine 1 also logged 3 entries averaging 20.0, while Machine 3 (2 entries) and Machine 8 (1 entry) both averaged 20.0, and Machine 2 (2 entries) averaged 19.0. Explicit UtilizationPercent metrics requested by the user were not available in the query schema, making CapacityId the primary capacity metric available.



Data preview — CapacityAnalysis (showing 5 of 5 points/rows)

MachineId	avg_CapacityId	record_count
6	23	3
8	20	1
1	20	3
3	20	2
2	19	2

Shift Production, Scrap, Rejection, and Telemetry Data Gap Analysis

The datasets requested for shift-wise production (WeighingMachine), scrap by material (Scrap), and rejection analysis by reason (Rejected) were not populated in the retrieved SQL query response. Additionally, InfluxDB telemetry streams for all 12 monitored machines (BM-01 through BM-12) returned NO DATA across all historical evaluation windows up to 8,760 hours. This reveals a critical telemetry logging and table configuration gap in the MES reporting setup that warrants immediate investigation by the systems engineering team.

Appendix — Data Quality

Table	Records	Completeness	Outliers	Score
cameras	3	100.0%	0	Good
camera_status_logs	50	100.0%	0	Good
attendances	1	100.0%	0	Good
detection_assignments	1	100.0%	0	Good
employees	7	100.0%	0	Good
CapacityAnalysis	5	100.0%	0	Good
MachineMaster	25	89.3%	0	Fair
WorkOrder	2	100.0%	0	Good
WorkOrderLog	1	75.0%	0	Needs review
MachineMaterialMapping	8	100.0%	0	Good
Machine	10	90.0%	0	Fair
MachineFGMapping	39	68.6%	0	Needs review

LLM Usage & Cost

Model	Input tokens	Output tokens	Total tokens	Estimated cost
gemini/gemini-3.5-flash	5,731	2,510	8,241	$0.0137

Aggregation Fallbacks

[mes.Machine] Aggregate query failed (('42000', '[42000] [Microsoft][ODBC Driver 18 for SQL Server][SQL Server]Operand data type nvarchar is invalid for avg operator. (8117) (SQLExecDirectW)')) - served raw rows instead.

InfluxDB Measurements With No Data

[influxdb.BM-01] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-02] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-03] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-04] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-05] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-06] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-07] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows

[influxdb.BM-08] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-09] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-10] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-11] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-12] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows